ITELEC 4100
Lab # 2

INTRO TO AJAX

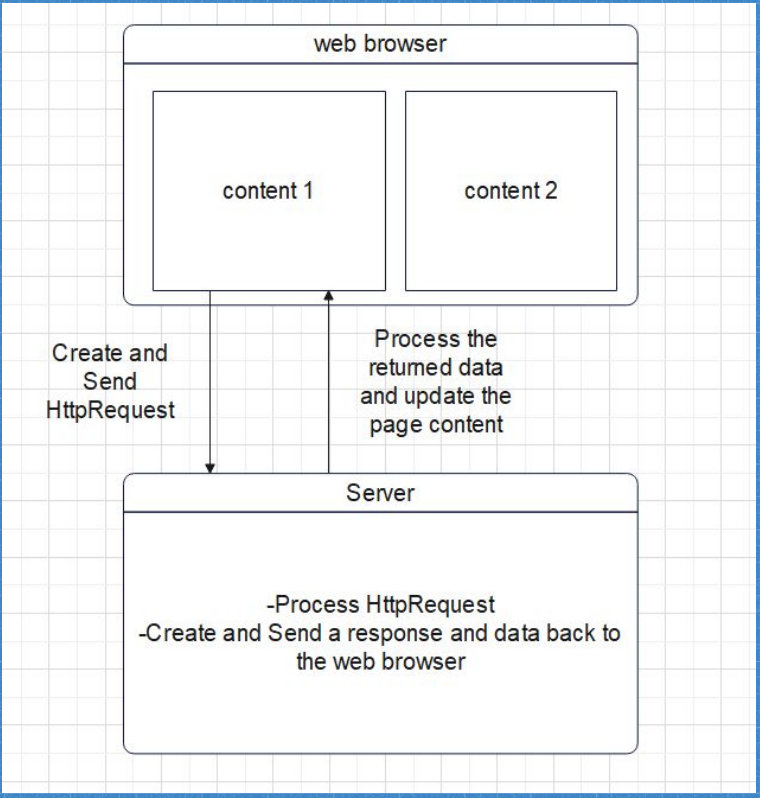
1

INTRODUCTION TO AJAX

What is AJAX?

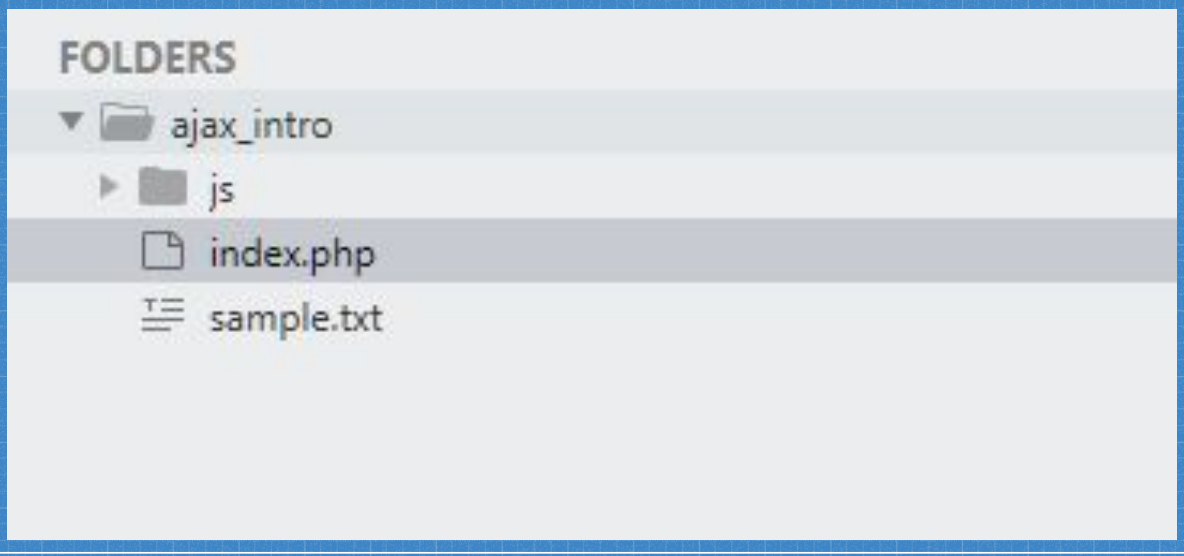
- AJAX or Asynchronous JavaScript and XML enable us to load data from the server without refreshing the whole web page

AJAX Concept



AJAX using Vanilla JavaScript (exercise)

- Create new project inside htdocs



AJAX using Vanilla JavaScript (exercise)

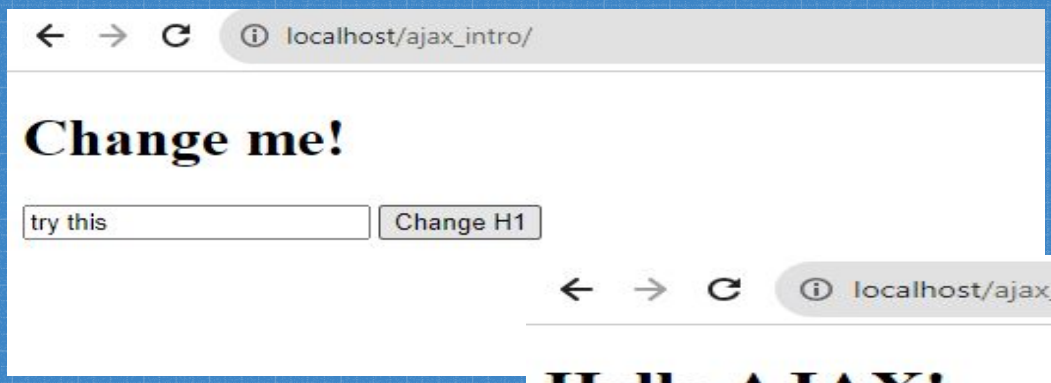
- sample.txt

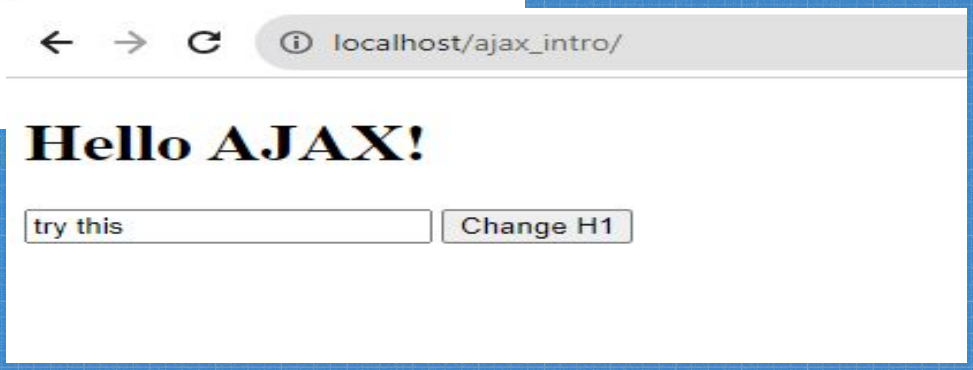
index.php

sample.txt | index.php

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title></title>
</head>
<body>
    <h1 id="demo">Change me!</h1>
    <input type="text">
    <button onclick="changeH1()">Change H1</button>
</body>
<script type="text/javascript">
    function changeH1(){
        var xhttp = new XMLHttpRequest();
        xhttp.open("GET","sample.txt",true);
        xhttp.send();
        xhttp.onreadystatechange = function(){
        if(this.readyState == 4 && this.status == 200){
                document.getElementById("demo").innerHTML = this.responseText;
            }
            console.log(this);
        };
    }
</script>
</html>
```


AJAX using Vanilla JavaScript (exercise)

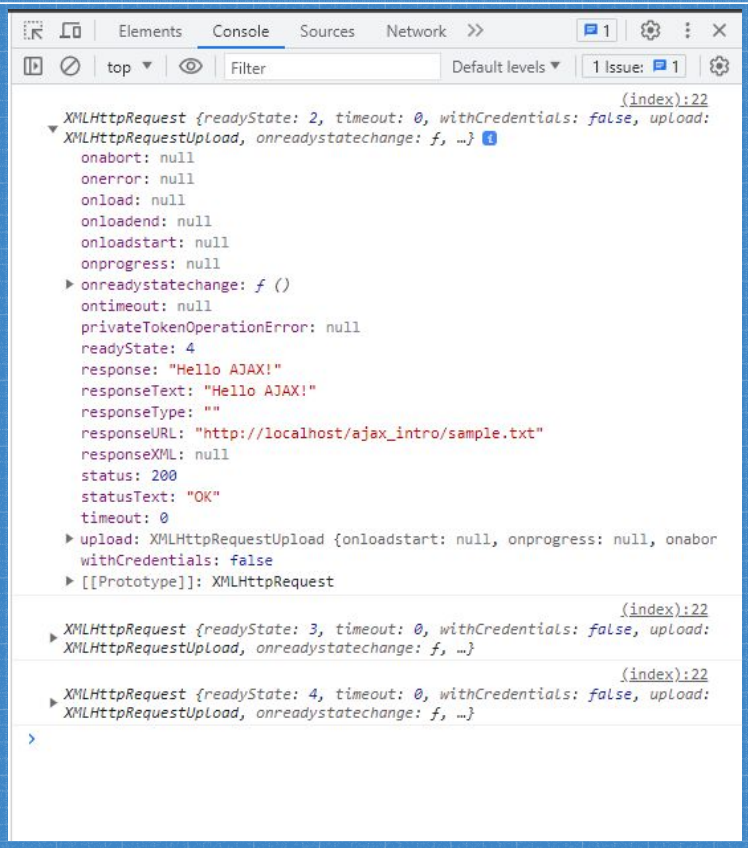




AJAX using Vanilla JavaScript (exercise)

cntrl+shift+i

//console.log()

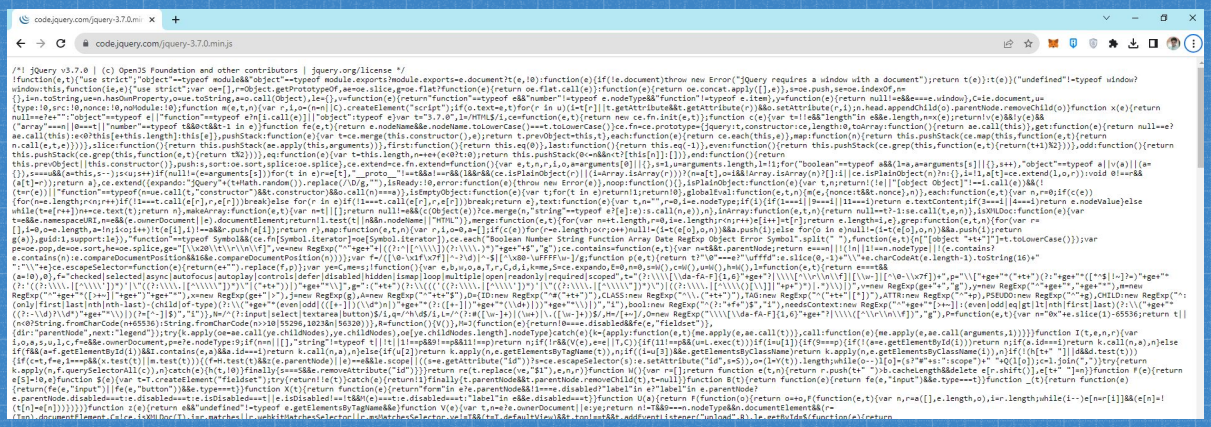


2

JQuery

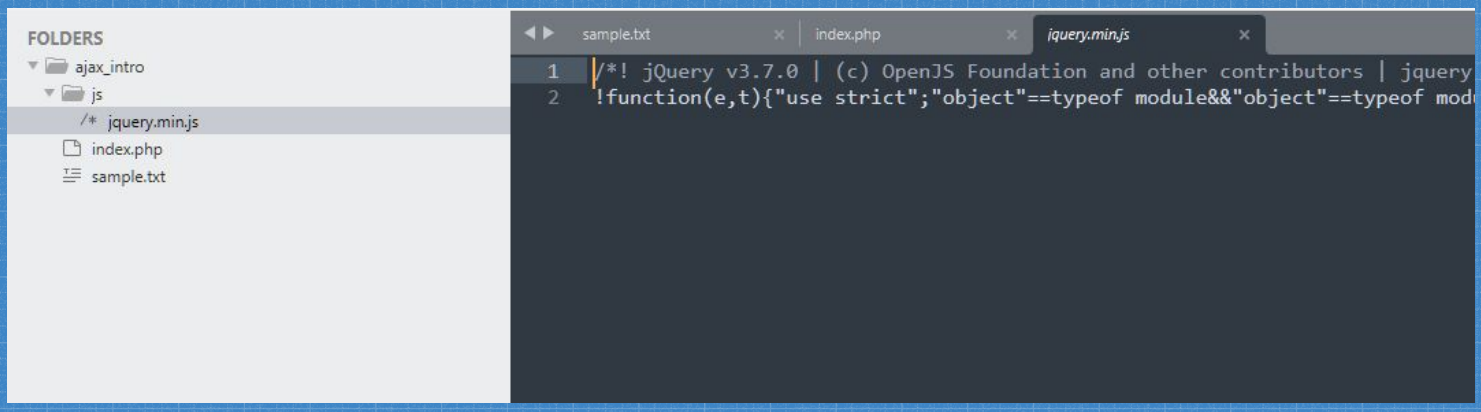
JQuery

- LINK:
  - https://code.jquery.com/jquery-3.7.0.min.js



JQuery

- Copy and paste code in js/jquery.min.js



index.php

index.php | sample.txt | jquery.min.js

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title></title>
    <script type="text/javascript" src="js/jquery.min.js"></script>
</head>
```


index.php

```
<script type="text/javascript">
    function changeH1(){
/*      var xhttp = new XMLHttpRequest();
        xhttp.open("GET","sample.txt",true);
        xhttp.send();
        xhttp.onreadystatechange = function(){
        if(this.readyState == 4 && this.status == 200){
                document.getElementById("demo").innerHTML = this.responseText;
            }
            console.log(this);
        };
*/
            $.get('sample.txt', function(data) {
                $('#demo').html(data);
            }, 'text');
    }


</script>
```

3

CRUD Function using JQuery

Database (oop)

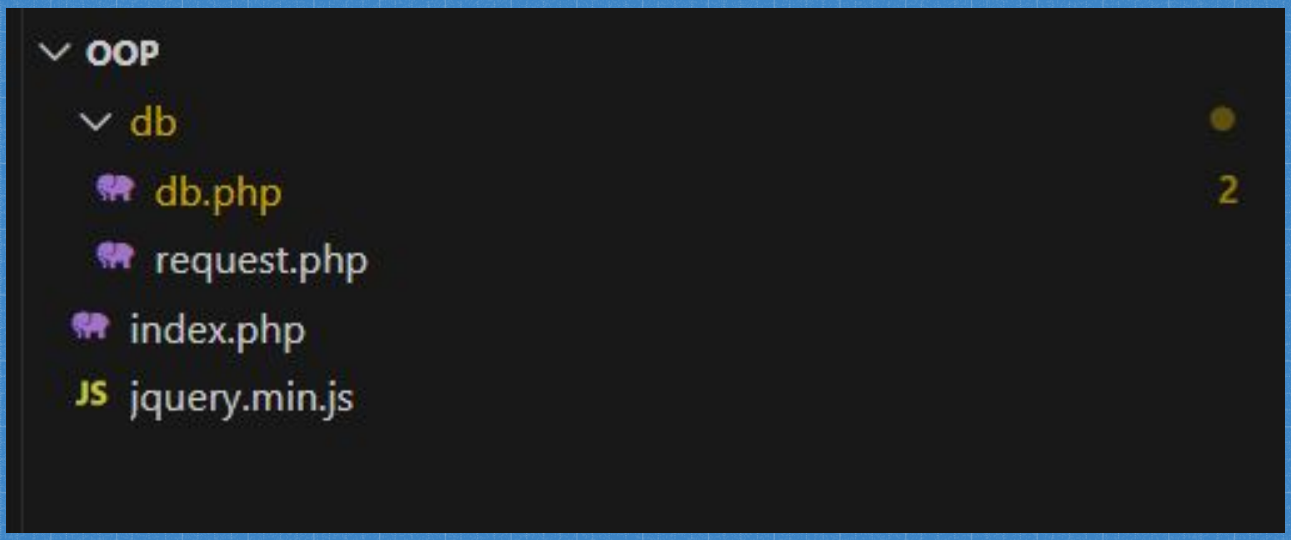
tbl_students

Table structure | Relation view

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
☐	1	**id**	int(11)			No	*None*		AUTO_INCREMENT	Change	Drop	More
☐	2	**full_name**	varchar(100)	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	3	**email**	varchar(100)	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	4	**course_year_section**	varchar(100)	utf8mb4_general_ci		No	*None*			Change	Drop	More

AJAX with JQuery

- Use the activity “oop”



index.php

```php
<?php
    require "db/db.php";
    $mydb = new myDB();
    $mydb->select('tbl_students');
    $student_data = $mydb->res;
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="initial-scale=1.0">
    <title>Document</title>
    <script type="text/javascript" src="jquery.min.js"></script>
</head>
<body>
    <form id="addStudentForm" method="post" action="db/request.php">…
    </form>
    <table>
        <thead>…
        </thead>
        <tbody id="tBodyStudent">

        </tbody>
    </table>
</body>
<script type="text/javascript">
    $(document).ready(function(){
        loadStudents();
    })
    function loadStudents(){…

    $("#addStudentForm").on("submit",function(e){…
</script>
</html>
```

index.php // form

```
<form id="addStudentFomr" method="post" action="db/request.php">
    <input type="text" name="full_name">
    <input type="text" name="email">
    <input type="text" name="course_year_section">
    <input type="submit" name="add_student" value="ADD">
</form>
```

index.php // table body

```
<tbody id="tBodyStudent">

</tbody>
table>
```

index.php

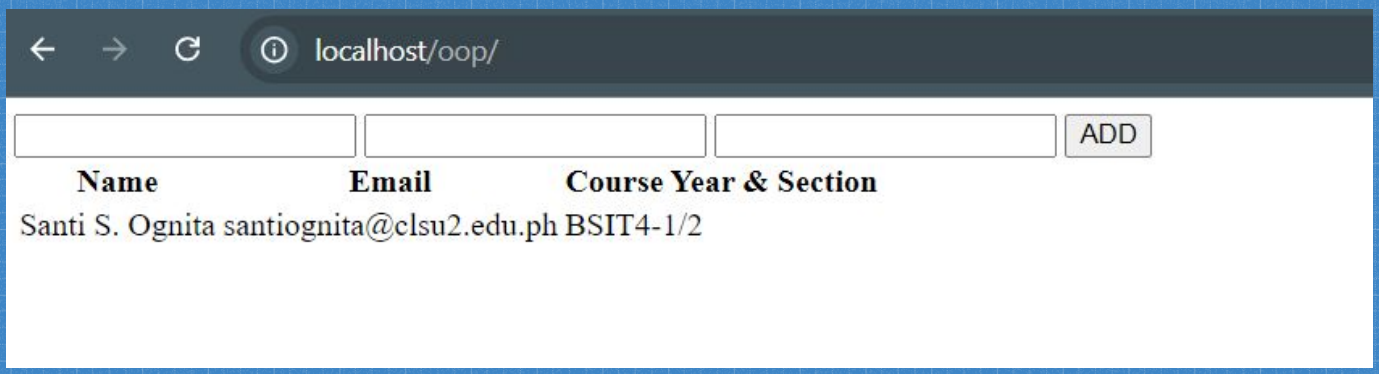
// script

```javascript
function loadStudents(){
    $.ajax({
        url:"db/request.php",
        method:"POST",
        data:{
            "get_student":true,
        },
        success:function(result){
            //console.log(result)
            var tBody = ``;
            var cnt = 1;
            var datas = JSON.parse(result);
            datas.forEach(function(data){
                tBody +=`<tr>`;
                    tBody +=`<td>${data['full_name']}</td>`;
                    tBody +=`<td>${data['email']}</td>`;
                    tBody +=`<td>${data['course_year_section']}</td>`;
                tBody +=`</tr>`;
            });
            $('#tBodyStudent').html(tBody);
        },
        error:function(error){
            //console.log(error);
            alert("Something went wrong!");
        }
    })
}
```

request.php

```php
<?php

require "db.php";
$mydb = new myDB();

if(isset($_POST['add_student'])){ ...
}

if(isset($_POST['get_student'])){
    $mydb->select('tbl_students');
    $datas = [];
    while($row = $mydb->res->fetch_assoc()){
        array_push($datas,$row);
    }
    echo json_encode($datas);
}

?>
```

OUTPUT



index.php

// script

```
$("#addStudentForm").on("submit",function(e){
    e.preventDefault();
    var datas = $(this).serializeArray();
    var data_array = {};
    $.map(datas,function(data){
        data_array[data['name']]=data['value'];
    });
    console.log(datas);
    console.log(data_array);
})
```

OUTPUT

//console

```
▼ (3) [{…}, {…}, {…}] i                                          oop/:68
  ▶ 0: {name: 'full_name', value: 'a'}
  ▶ 1: {name: 'email', value: 'a'}
  ▶ 2: {name: 'course_year_section', value: 'a'}
    length: 3
  ▶ [[Prototype]]: Array(0)

▼ {full_name: 'a', email: 'a', course_year_section: 'a'} i          oop/:69
    course_year_section: "a"
    email: "a"
    full_name: "a"
  ▶ [[Prototype]]: Object
>
```

index.php

// script

```
$("#addStudentForm").on("submit",function(e){
    e.preventDefault();
    var datas = $(this).serializeArray();
    var data_array = {};
    $.map(datas,function(data){
        data_array[data['name']]=data['value'];
    });
    //console.log(datas);
    //console.log(data_array);
    $.ajax({
        url:"db/request.php",
        method:"POST",
        data:{
            'add_student':true,
            ...data_array,
        },
        success:function(result){
            //console.log(result);
            loadStudents();
        },
        error:function(error){
            //console.log(error);
            alert("Something went wrong!");
        }
    })
})
```

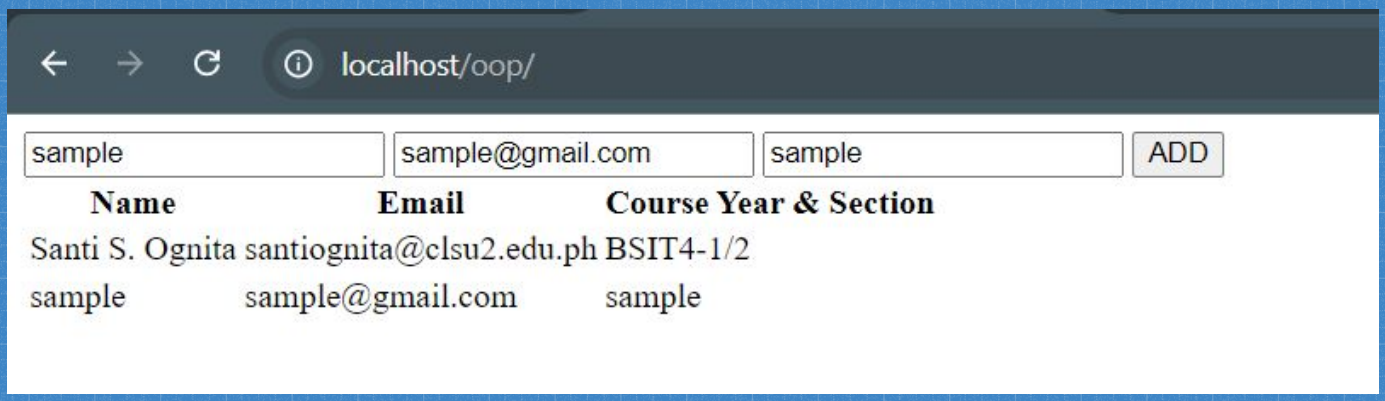
request.php

```php
if(isset($_POST['add_student'])){
    unset($_POST['add_student']);
    $mydb->insert('tbl_students',$_POST);
}
```

db.php

```php
1 reference | 0 overrides
public function insert($table,$data){
    try{ ...
    }catch(Exception $e){
        echo ("Error while inserting data! <br>".$e);
    }
}

2 references | 0 overrides
public function select($table,$row="*",$where=NULL){
    try{ ...
    }catch(Exception $e){
        echo ("Error requesting data! <br>".$e);
    }
}
```

OUTPUT



Activity 2

1. ADD UPDATE FUNCTION
2. ADD DELETE FUNCTION
3. ADD SEARCH FUNCTION ON TABLE DATA (DON'T USE DATATABLES PLUG-IN or any PLUGINS)
4. ADD DESIGN (BOOTSTRAP OR CUSTOM CSS)